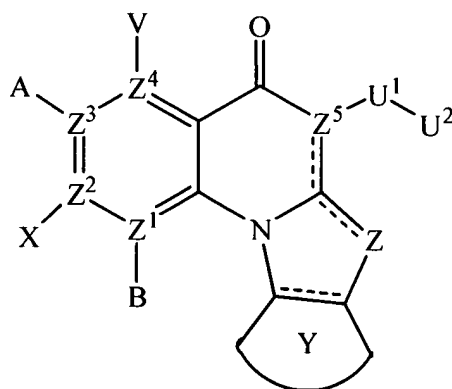


WHAT IS CLAIMED IS:

1. A method of treating an autoimmune disease in a subject, comprising:
administering to the subject a therapeutically effective amount of a compound of
Formula (I),



Formula (I)

or a pharmaceutically acceptable salt or ester thereof;

wherein ----- indicates an optionally unsaturated bond;

each of B, X, A or V is absent if Z^1 , Z^2 , Z^3 and Z^4 , respectively, is N; and

each of B, X, A and V is independently H, halo, azido, —CN, —CF₃, —CONR¹R², —NR¹R², —SR², —OR², —R³, —W, —L-W, —W⁰, —L-N(R) —W⁰, A² or A³, when each of Z^1 , Z^2 , Z^3 and Z^4 , respectively, is C;

Z is O, S, CR⁴₂, NR⁴CR⁴, CR⁴NR⁴, CR⁴, NR⁴ or N;

each of Z^1 , Z^2 , Z^3 and Z^4 is independently C or N, provided any three N are non-adjacent;

Z^5 is C; or Z^5 may be N when Z is N;

Y is an optionally substituted 5-6 membered carbocyclic or heterocyclic ring;

U¹ is —C(=T)N(R)—, —C(=T)N(R)O—, —C(=T)—, —SO₂N(R)—, —SO₂N(R)N(R⁰)—, —SO₂—, or —SO₃—, where T is O, S, or NH; or U¹ may be a bond when Z^5 is N or U² is H;

U² is H, or C3-C7 cycloalkyl, C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl or C2-C10 heteroalkenyl group, each of which may be optionally substituted with one or more

halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring; or U^2 is -W, -L-W, -L-N(R)- W^0 , A^2 or A^3 ;

in each -NR¹R², R¹ and R² together with N may form an optionally substituted azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member;

R¹ is H or C1-C6 alkyl, optionally substituted with one or more halogens, or =O;

R² is H, or C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl, or C2-C10 heteroalkenyl, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring;

R³ is an optionally substituted C1-C10 alkyl, C2-C10 alkenyl, C5-C10 aryl, or C6-C12 arylalkyl, or a heteroform of one of these, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-6 membered carbocyclic or heterocyclic ring;

each R⁴ is independently H, or C1-C6 alkyl; or R⁴ may be -W, -L-W or -L-N(R)- W^0 ;

each R and R⁰ is independently H or C1-C6 alkyl;

L is a C1-C10 alkylene, C1-C10 heteroalkylene, C2-C10 alkenylene or C2-C10 heteroalkenylene linker, each of which may be optionally substituted with one or more substituents selected from the group consisting of halogen, oxo (=O), or C1-C6 alkyl;

W is an optionally substituted 4-7 membered azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member;

W^0 is an optionally substituted 3-4 membered carbocyclic ring, or a C1-C6 alkyl group substituted with from 1 to 4 fluorine atoms;

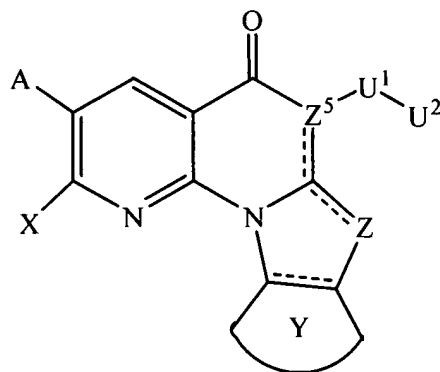
provided one of U^2 , V, A, X and B is a secondary amine A^2 or a tertiary amine A^3 , wherein

the secondary amine A^2 is -NH- W^0 , and

the tertiary amine A^3 is a fully saturated and optionally substituted six-membered or seven-membered azacyclic ring optionally containing an additional heteroatom selected from N, O or S as a ring member, or the tertiary amine A^3 is a partially unsaturated or aromatic optionally substituted five-membered azacyclic ring, optionally containing an additional heteroatom selected from N, O or S as a ring member;

thereby treating the autoimmune disease in the subject.

2. A method of treating an autoimmune disease in a subject, comprising:
administering to the subject a therapeutically effective amount of a compound of
Formula (II),



Formula (II)

or a pharmaceutically acceptable salt or ester thereof;

wherein ----- indicates an optionally unsaturated bond;

each of A and X is independently H, halo, azido, -CN, -CF₃, -CONR¹R², -NR¹R², -SR², -OR², -R³, -W, -L-W, -W⁰, -L-N(R)-W⁰, A² or A³;

Z is O, S, CR⁴₂, NR⁴CR⁴, CR⁴NR⁴ or NR⁴;

Y is an optionally substituted 5-6 membered carbocyclic or heterocyclic ring;

U¹ is -C(=T)N(R)-, -C(=T)N(R)O-, -C(=T)-, -SO₂N(R)-, -SO₂N(R)N(R⁰)-, -SO₂-, or -SO₃-, where T is O, S, or NH; or U¹ may be a bond when U² is H;

U² is H, or C3-C7 cycloalkyl, C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl or C2-C10 heteroalkenyl group, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring; or U² is -W, -L-W, -L-N(R)-W⁰, A² or A³;

in each -NR¹R², R¹ and R² together with N may form an optionally substituted azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member;

R¹ is H or C1-C6 alkyl, optionally substituted with one or more halogens, or =O;

R² is H, or C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl, or C2-C10 heteroalkenyl, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring;

R³ is an optionally substituted C1-C10 alkyl, C2-C10 alkenyl, C5-C10 aryl, or C6-C12 arylalkyl, or a heteroform of one of these, each of which may be optionally substituted

with one or more halogens, =O, or an optionally substituted 3-6 membered carbocyclic or heterocyclic ring;

each R^4 is independently H, or C1-C6 alkyl; or R^4 may be -W, -L-W or -L-N(R)-W⁰;

each R and R^0 is independently H or C1-C6 alkyl;

L is a C1-C10 alkylene, C1-C10 heteroalkylene, C2-C10 alkenylene or C2-C10 heteroalkenylene linker, each of which may be optionally substituted with one or more substituents selected from the group consisting of halogen, oxo (=O), or C1-C6 alkyl;

W is an optionally substituted 4-7 membered azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member;

W⁰ is an optionally substituted 3-4 membered carbocyclic ring, or a C1-C6 alkyl group substituted with from 1 to 4 fluorine atoms;

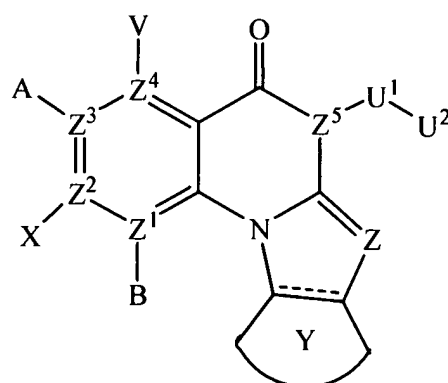
provided that one of U², A, and X is a secondary amine A² or a tertiary amine A³, wherein

the secondary amine A² is -NH-W⁰, and

the tertiary amine A³ is a fully saturated and optionally substituted six-membered or seven-membered azacyclic ring optionally containing an additional heteroatom selected from N, O or S as a ring member, or the tertiary amine A³ is a partially unsaturated or aromatic optionally substituted five-membered azacyclic ring optionally containing an additional heteroatom selected from N, O or S as a ring member;

thereby treating the autoimmune disease in the subject.

3. A method of treating an autoimmune disease in a subject, comprising:
administering to the subject a therapeutically effective amount of a compound of Formula (III),



Formula (III)

or a pharmaceutically acceptable salt or ester thereof;

wherein ---- indicates an optionally unsaturated bond; and

each of B, X, A or V is absent if Z^1 , Z^2 , Z^3 and Z^4 , respectively, is N; and

each of B, X, A and V is independently H, halo, azido, -CN, -CF₃, -CONR¹R², -NR¹R², -SR², -OR², -R³, -W, -L-W, -W⁰, -L-N(R)-W⁰, A² or A³, when each of Z^1 , Z^2 , Z^3 and Z^4 , respectively, is C;

each of Z^1 , Z^2 , Z^3 and Z^4 is independently C or N, provided any three N are non-adjacent;

Y is an optionally substituted 5-6 membered carbocyclic or heterocyclic ring;

U¹ is -C(=T)N(R)-, -C(=T)N(R)O-, -C(=T)-, -SO₂N(R)-, -SO₂N(R)N(R⁰)-, -SO₂-, or -SO₃-, where T is O, S, or NH; or U¹ may be a bond when Z^5 is N or U² is H;

U² is H, or C3-C7 cycloalkyl, C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl or C2-C10 heteroalkenyl group, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring; or U² is -W, -L-W, -L-N(R)-W⁰, A² or A³;

in each -NR¹R², R¹ and R² together with N may form an optionally substituted azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member;

R¹ is H or C1-C6 alkyl, optionally substituted with one or more halogens, or =O;

R² is H, or C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl, or C2-C10 heteroalkenyl, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring;

R³ is an optionally substituted C1-C10 alkyl, C2-C10 alkenyl, C5-C10 aryl, or C6-C12 arylalkyl, or a heteroform of one of these, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-6 membered carbocyclic or heterocyclic ring;

each R⁴ is independently H, or C1-C6 alkyl; or R⁴ may be -W, -L-W or -L-N(R)-W⁰;

each R and R⁰ is independently H or C1-C6 alkyl;

L is a C1-C10 alkylene, C1-C10 heteroalkylene, C2-C10 alkenylene or C2-C10 heteroalkenylene linker, each of which may be optionally substituted with one or more substituents selected from the group consisting of halogen, oxo (=O), or C1-C6 alkyl;

W is an optionally substituted 4-7 membered azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member;

W⁰ is an optionally substituted 3-4 membered carbocyclic ring, or a C1-C6 alkyl group substituted with from 1 to 4 fluorine atoms;

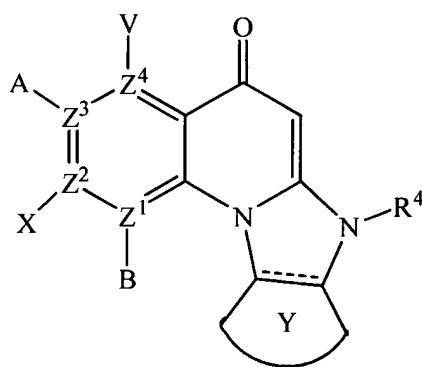
provided that one of U², V, A, X and B is a secondary amine A² or a tertiary amine A³, wherein

the secondary amine A² is -NH-W⁰, and

the tertiary amine A³ is a fully saturated and optionally substituted six-membered or seven-membered azacyclic ring optionally containing an additional heteroatom selected from N, O or S as a ring member, or the tertiary amine A³ is a partially unsaturated or aromatic optionally substituted five-membered azacyclic ring, optionally containing an additional heteroatom selected from N, O or S as a ring member,

thereby treating the autoimmune disease in the subject.

4. A method of treating an autoimmune disease in a subject, comprising:
administering to the subject a therapeutically effective amount of a compound of Formula (IV),



Formula (IV)

or a pharmaceutically acceptable salt or ester thereof;

wherein ----- indicates an optionally unsaturated bond;

each of B, X, A or V is absent if Z¹, Z², Z³ and Z⁴ respectively, is N; and

each of B, X, A and V is independently H, halo, azido, -CN, -CF₃, -CONR¹R², —NR¹R², -SR², -OR², -R³, -W, -L-W, -W⁰, -L-N(R)-W⁰, A² or A³, when each of Z¹, Z², Z³ and Z⁴, respectively, is C;

each of Z^1 , Z^2 , Z^3 and Z^4 is independently C or N, provided any three N are non-adjacent;

Y is an optionally substituted 5-6 membered carbocyclic or heterocyclic ring;

in each NR^1R^2 , R^1 and R^2 together with N may form an optionally substituted azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member;

R^1 is H or C1-C6 alkyl, optionally substituted with one or more halogens, or =O;

R^2 is H, or C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl, or C2-C10 heteroalkenyl, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring;

R^3 is an optionally substituted C1-C10 alkyl, C2-C10 alkenyl, C5-C10 aryl, or C6-C12 arylalkyl, or a heteroform of one of these, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-6 membered carbocyclic or heterocyclic ring;

R^4 is -W, -L-W or -L-N(R)-W⁰; and

each R is independently H or C1-C6 alkyl;

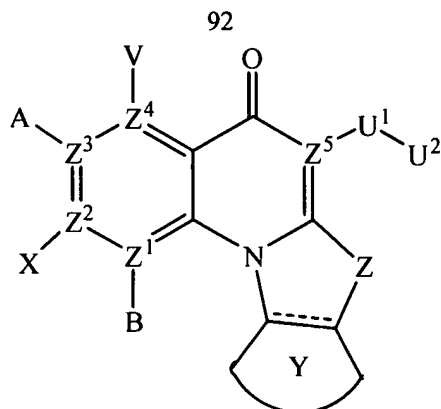
L is a C1-C10 alkylene, C1-C10 heteroalkylene, C2-C10 alkenylene or C2-C10 heteroalkenylene linker, each of which may be optionally substituted with one or more substituents selected from the group consisting of halogen, oxo (=O), or C1-C6 alkyl;

W is an optionally substituted 4-7 membered azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member; and

W⁰ is an optionally substituted 3-4 membered carbocyclic ring, or a C1-C6 alkyl group substituted with from 1 to 4 fluorine atoms,

thereby treating the autoimmune disease in the subject.

5. A method of treating an autoimmune disease in a subject, comprising:
administering to the subject a therapeutically effective amount of a compound of Formula (V),



Formula (V)

or a pharmaceutically acceptable salt or ester thereof;

---- indicates an optionally unsaturated bond;

A and V independently are H, halo, azido, -CN, -CF₃, -CONR¹R², -NR¹R², -SR², -OR², -R³, -W, -L-W, -W⁰, -L-N(R)-W⁰, A² or A³;

Z is O, S, CR⁴₂, NR⁴CR⁴, CR⁴NR⁴ or NR⁴;

Y is an optionally substituted 5-6 membered carbocyclic or heterocyclic ring;

U¹ is -C(=T)N(R)-, -C(=T)N(R)O-, -C(=T)-, -SO₂N(R)-, -SO₂N(R)N(R⁰)-, -SO₂-, or -SO₃-, where T is O, S, or NH; or U¹ may be a bond when U² is H;

U² is H, or C3-C7 cycloalkyl, C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl or C2-C10 heteroalkenyl group, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring; or U² is -W, -L-W or -L-N(R)-W⁰, A² or A³;

in each -NR¹R², R¹ and R² together with N may form an optionally substituted azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member;

R¹ is H or C1-C6 alkyl, optionally substituted with one or more halogens, or =O;

R² is H, or C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl, or C2-C10 heteroalkenyl, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring;

R³ is an optionally substituted C1-C10 alkyl, C2-C10 alkenyl, C5-C10 aryl, or C6-C12 arylalkyl, or a heteroform of one of these, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-6 membered carbocyclic or heterocyclic ring;

each R⁴ is independently H, or C1-C6 alkyl; or R⁴ may be -W, -L-W or -L-N(R)-W⁰;

each R and R⁰ is independently H or C1-C6 alkyl;

L is a C1-C10 alkylene, C1-C10 heteroalkylene, C2-C10 alkenylene or C2-C10 heteroalkenylene linker, each of which may be optionally substituted with one or more substituents selected from the group consisting of halogen, oxo (=O), or C1-C6 alkyl;

W is an optionally substituted 4-7 membered azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member;

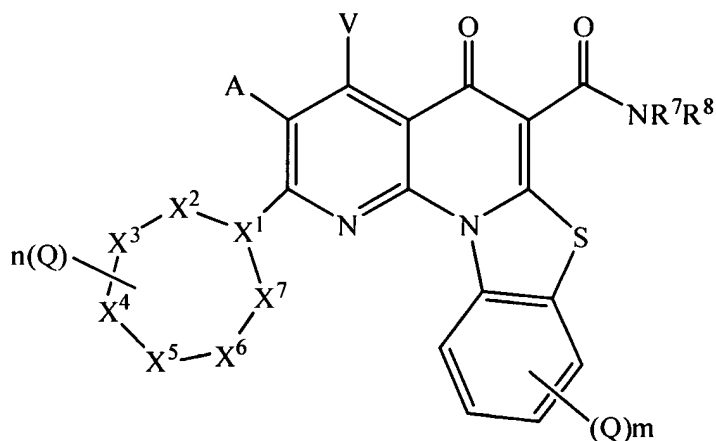
W⁰ is an optionally substituted 3-4 membered carbocyclic ring, or a C1-C6 alkyl group substituted with from 1 to 4 fluorine atoms;

provided one of U², A, and V is a secondary amine A² or a tertiary amine A³, wherein the secondary amine A² is -NH-W⁰, and

the tertiary amine A³ is a fully saturated and optionally substituted six-membered or seven-membered azacyclic ring optionally containing an additional heteroatom selected from N, O or S as a ring member, or the tertiary amine A³ is a partially unsaturated or aromatic optionally substituted five-membered azacyclic ring optionally containing an additional heteroatom selected from N, O or S as a ring member,

thereby treating the autoimmune disease in the subject.

6. A method of treating an autoimmune disease in a subject, comprising:
administering to the subject a therapeutically effective amount of a compound of Formula (VI),



Formula (VI)

or a pharmaceutically acceptable salt or ester thereof;

wherein:

X¹ is CH or N;

X^2, X^3, X^4, X^5, X^6 and X^7 independently are NR^4 , CH_2 , CHQ or $C(Q)_2$, provided that: (i) zero, one or two of X^2, X^3, X^4, X^5, X^6 and X^7 are NR^4 ; (ii) when X^1 is N, both of X^2 and X^7 are not NR^4 ; (iii) when X^1 is N, X^3 and X^6 are not NR^4 ; and (iv) when X^1 is CH and two of X^2, X^3, X^4, X^5, X^6 and X^7 are NR^4 , the two NR^4 are located at adjacent ring positions or are separated by two or more other ring positions;

A and V independently are H, halo, azido, -CN, -CF₃, -CONR¹R², -NR¹R², -SR², -OR², -R³, -W, -L-W, -W⁰, or -L-N(R)-W⁰;

each Q is independently halo, azido, -CN, -CF₃, -CONR¹R², -NR¹R², -SR², -OR², -R³, -W, -L-W, -W⁰, or -L-N(R)-W⁰;

in each —NR¹R², R¹ and R² together with N may form an optionally substituted azacyclic ring, optionally containing one additional heteroatom selected from N, O and S as a ring member;

R¹ is H or C1-C6 alkyl, optionally substituted with one or more halogens, or =O;

R is H, or C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl, or C2-C10 heteroalkenyl, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring;

R³ is an optionally substituted C1-C10 alkyl, C2-C10 alkenyl, C5-C10 aryl, or C6-C12 arylalkyl, or a heteroform of one of these, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-6 membered carbocyclic or heterocyclic ring;

each R⁴ is independently H, or C1-C6 alkyl; or R⁴ may be -W, -L-W or -L-N(R)-W⁰;

each R is independently H or C1-C6 alkyl;

R⁷ is H and R⁸ is C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl, or C2-C10 heteroalkenyl, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring; or in —NR⁷R⁸, R⁷ and R⁸ together with N may form an optionally substituted azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member;

m is 0, 1, 2, 3 or 4;

n is 0, 1, 2, 3, 4, or 5;

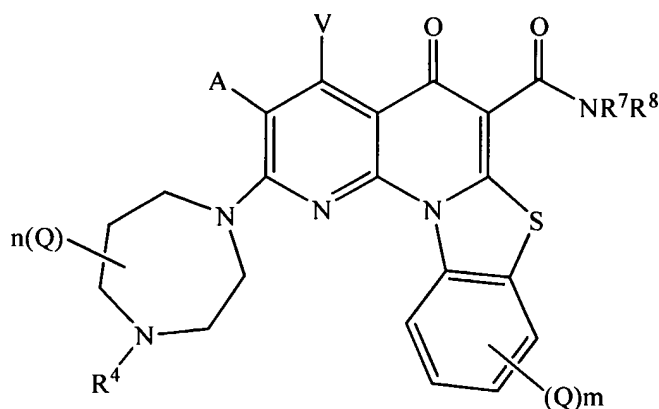
L is a C1-C10 alkylene, C1-C10 heteroalkylene, C2-C10 alkenylene or C2-C10 heteroalkenylene linker, each of which may be optionally substituted with one or more substituents selected from the group consisting of halogen, oxo (=O), or C1-C6 alkyl;

W is an optionally substituted 4-7 membered azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member; and

W⁰ is an optionally substituted 3-4 membered carbocyclic ring, or a C1-C6 alkyl group substituted with from 1 to 4 fluorine atoms,

thereby treating the autoimmune disease in the subject.

7. A method of treating an autoimmune disease in a subject, comprising:
administering to the subject a therapeutically effective amount of a compound of Formula (VIII),



Formula (VIII)

or a pharmaceutically acceptable salt or ester thereof;

wherein:

A and V independently are H, halo, azido, -CN, -CF₃, -CONR¹R², -NR¹R², -SR², -OR², -R³, -W, -L-W, -W⁰, or -L-N(R)-W⁰;

each Q is independently halo, azido, -CN, -CF₃, -CONR¹R², -NR¹R², -SR², -OR², -R³, -W, -L-W, -W⁰, or -L-N(R)-W⁰;

in each —NR¹R², R¹ and R² together with N may form an optionally substituted azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member;

R¹ is H or C1-C6 alkyl, optionally substituted with one or more halogens, or =O;

R^2 is H, or C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl, or C2-C10 heteroalkenyl, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring;

R^3 is an optionally substituted C1-C10 alkyl, C2-C10 alkenyl, C5-C10 aryl, or C6-C12 arylalkyl, or a heteroform of one of these, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-6 membered carbocyclic or heterocyclic ring;

each R^4 is independently H, or C1-C6 alkyl; or R^4 may be -W, -L-W or -L-N(R)-W⁰;

each R is independently H or C1-C6 alkyl;

R^7 is H and R^8 is C1-C10 alkyl, C1-C10 heteroalkyl, C2-C10 alkenyl, or C2-C10 heteroalkenyl, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring; or in —NR⁷R⁸, R^7 and R^8 together with N may form an optionally substituted azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member;

m is 0, 1, 2, 3 or 4;

n is 0, 1, 2, 3, 4 or 5;

p is 0, 1, 2 or 3;

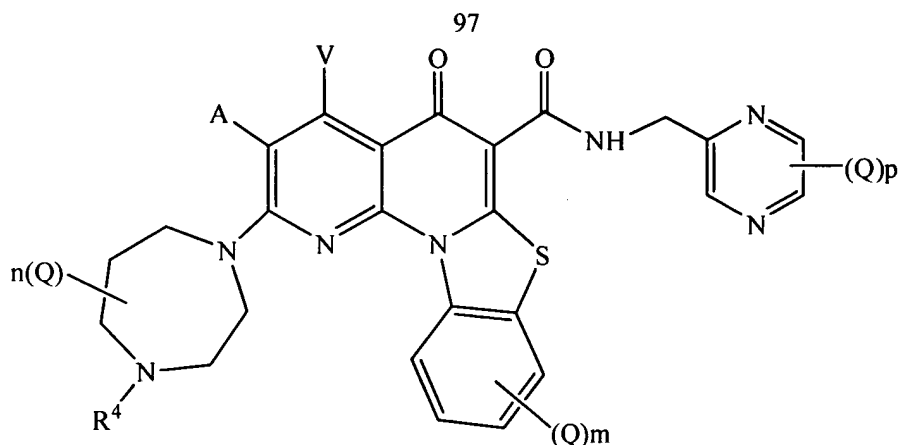
L is a C1-C10 alkylene, C1-C10 heteroalkylene, C2-C10 alkenylene or C2-C10 heteroalkenylene linker, each of which may be optionally substituted with one or more substituents selected from the group consisting of halogen, oxo (=O), or C1-C6 alkyl;

W is an optionally substituted 4-7 membered azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member; and

W⁰ is an optionally substituted 3-4 membered carbocyclic ring, or a C1-C6 alkyl group substituted with from 1 to 4 fluorine atoms,

thereby treating the autoimmune disease in the subject.

8. A method of treating an autoimmune disease in a subject, comprising:
administering to the subject a therapeutically effective amount of a compound of Formula (VII)



Formula (VII)

or a pharmaceutically acceptable salt or ester thereof;

A and V independently are H, halo, azido, -CN, -CF₃, -CONR¹R², -NR¹R², -SR², -OR², -R³, -W, -L-W, -W⁰, or -L-N(R)-W⁰;

each Q is independently halo, azido, -CN, -CF₃, -CONR¹R², -NR¹R², -SR², -OR², -R³, -W, -L-W, -W⁰, or -L-N(R)-W⁰;

in each —NR¹R², R¹ and R² together with N may form an optionally substituted azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member;

R¹ is H or C1-C6 alkyl, optionally substituted with one or more halogens, or =O;

R² is H, or C¹-C¹⁰ alkyl, C¹-C¹⁰ heteroalkyl, C²-C¹⁰ alkenyl, or C²-C¹⁰ heteroalkenyl, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-7 membered carbocyclic or heterocyclic ring;

R³ is an optionally substituted C1-C10 alkyl, C2-C10 alkenyl, C5-C10 aryl, or C6-C12 arylalkyl, or a heteroform of one of these, each of which may be optionally substituted with one or more halogens, =O, or an optionally substituted 3-6 membered carbocyclic or heterocyclic ring;

each R⁴ is independently H, or C1-C6 alkyl; or R⁴ may be -W, -L-W or -L-N(R)-W⁰;

each R is independently H or C1-C6 alkyl;

m is 0, 1, 2, 3 or 4;

n is 0, 1, 2, 3, 4 or 5;

p is 0, 1, 2 or 3;

L is a C1-C10 alkylene, C1-C10 heteroalkylene, C2-C10 alkenylene or C2-C10 heteroalkenylene linker, each of which may be optionally substituted with one or more substituents selected from the group consisting of halogen, oxo (=O), or C1-C6 alkyl;

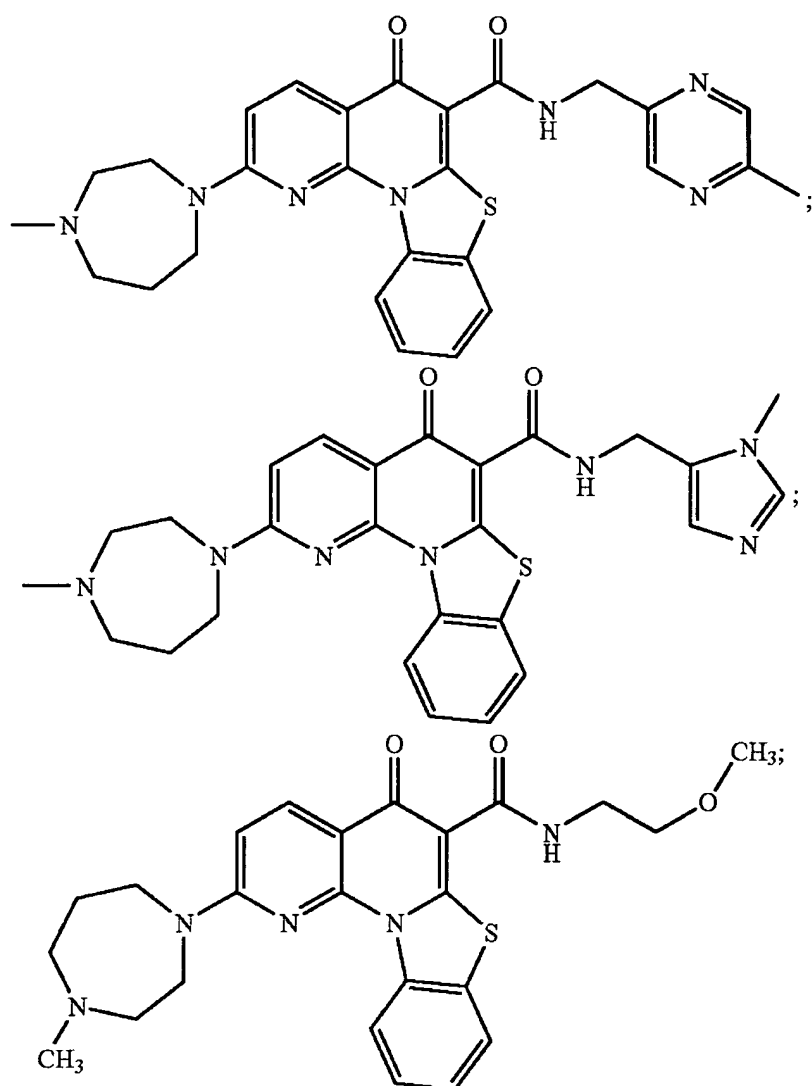
W is an optionally substituted 4-7 membered azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member; and

W⁰ is an optionally substituted 3-4 membered carbocyclic ring, or a C1-C6 alkyl group substituted with from 1 to 4 fluorine atoms,

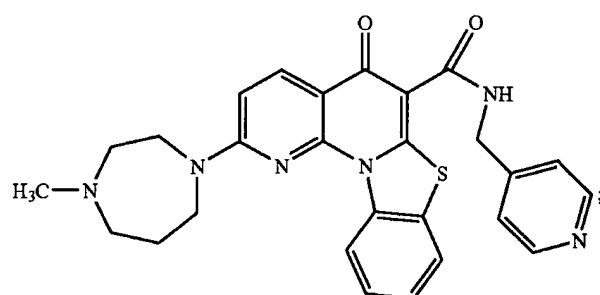
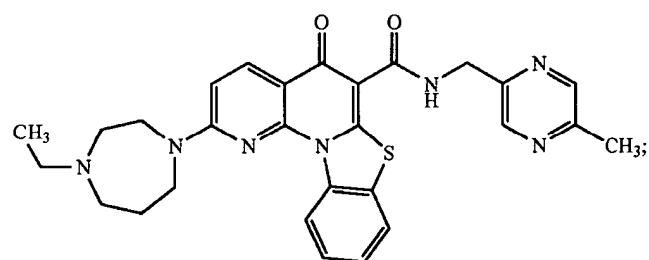
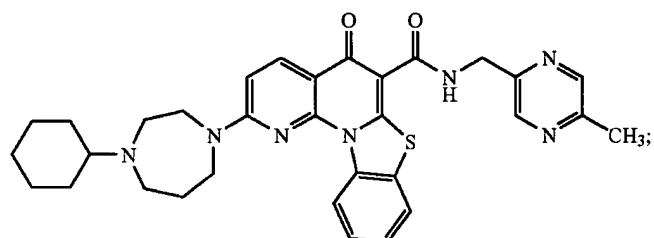
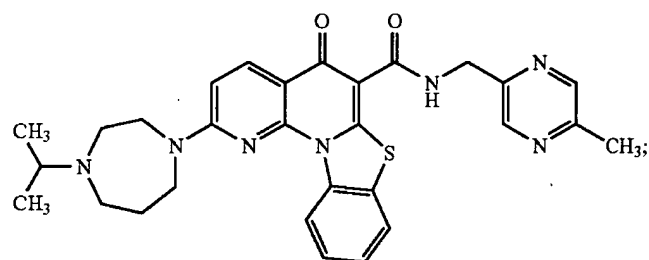
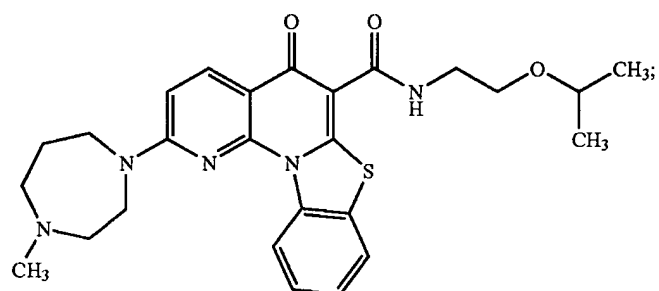
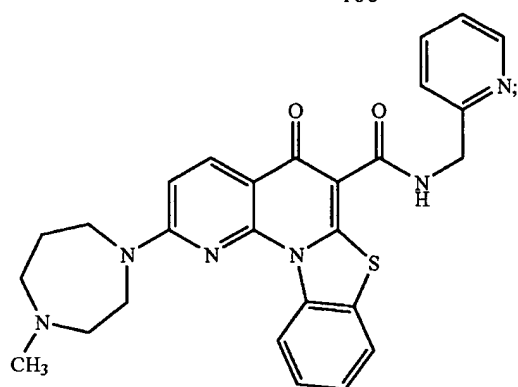
thereby treating the autoimmune disease in the subject.

9. A method of treating an autoimmune disease in a subject, comprising:
administering to the subject a therapeutically effective amount of a compound selected from the group consisting of:

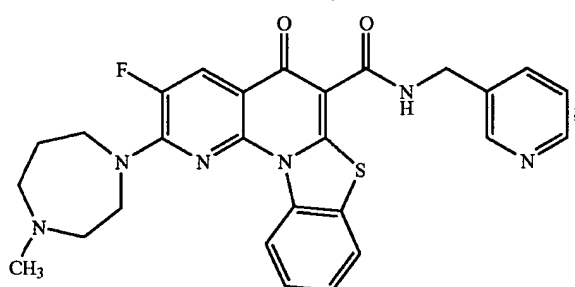
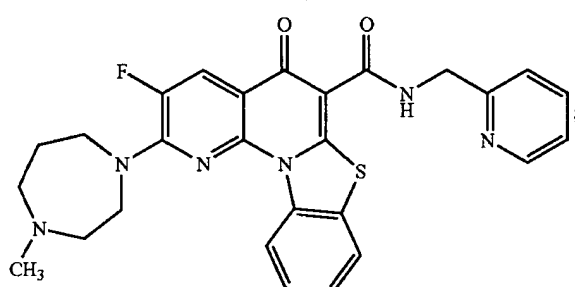
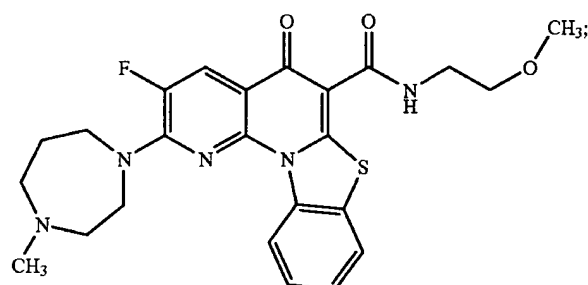
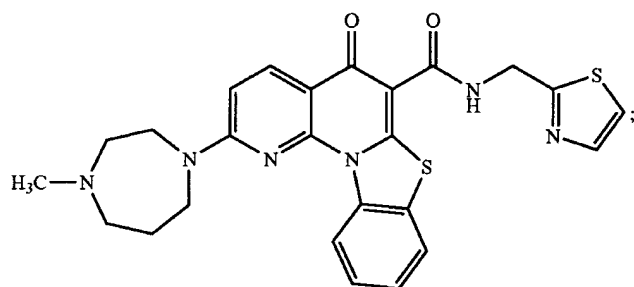
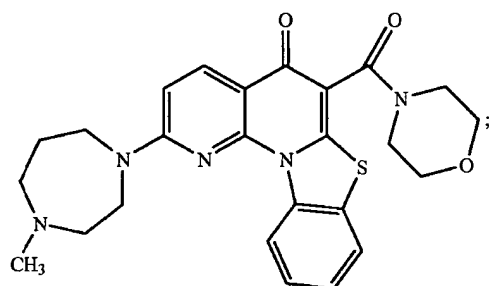
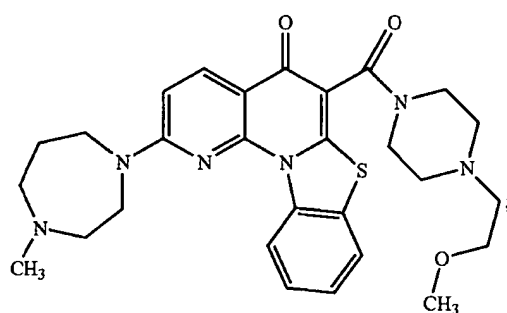
99



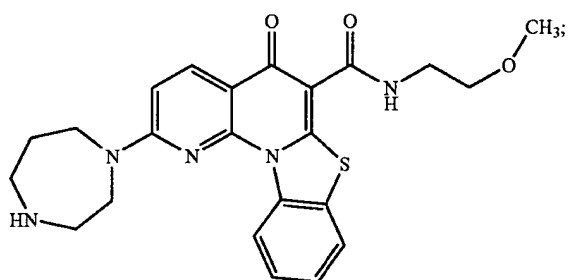
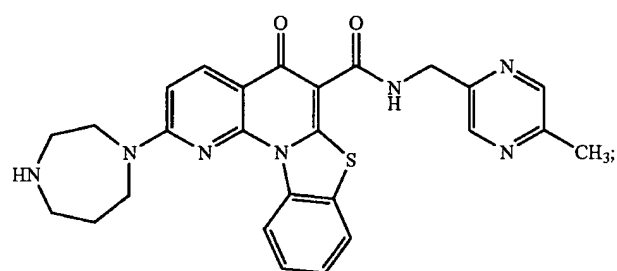
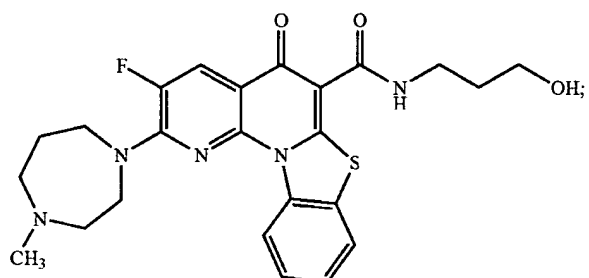
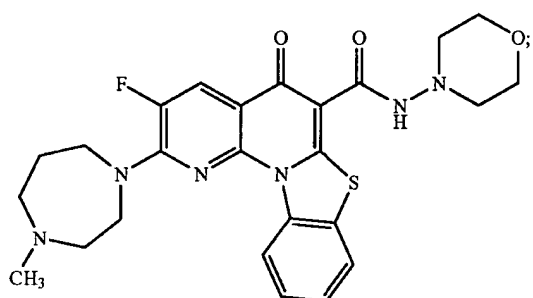
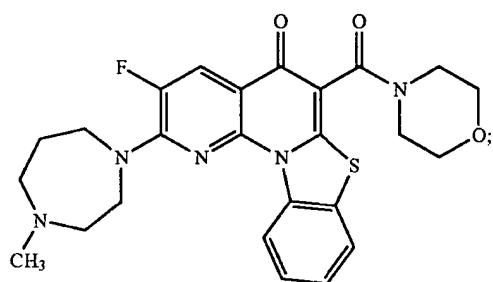
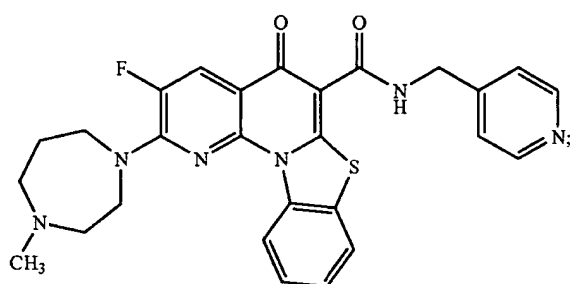
100

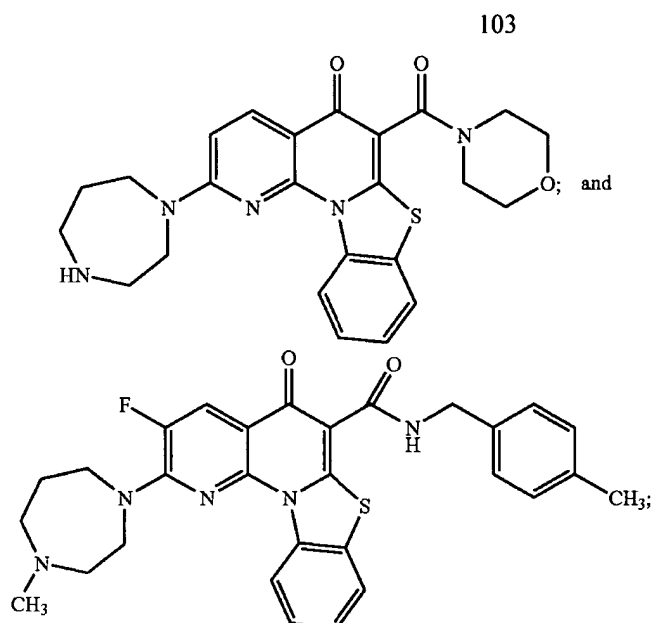


101



102





or a pharmaceutically acceptable salt or ester thereof,
thereby treating the autoimmune disease in the subject.

10. A method of monitoring treatment efficiency of a subject having an autoimmune disease, the method comprising:

(a) treating the subject with said compound according to the method of any of claims 1-9, and

(b) comparing a level of expression of least one gene involved in the RNA polymerase I pathway in a cell of the subject following said treating with said compound to a level of expression of said at least one gene in a cell of the subject prior to said treating the subject with said compound,

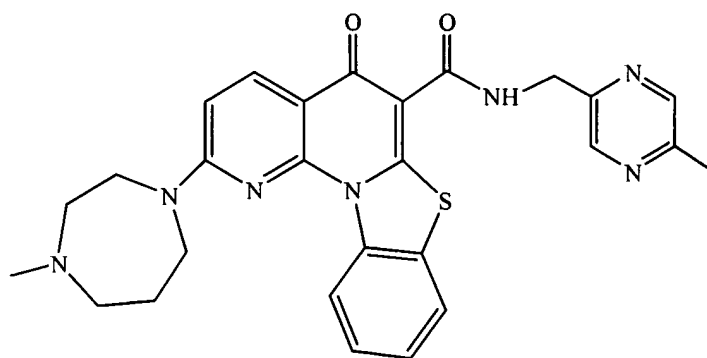
(i) wherein a decrease above a predetermined threshold in said level of expression of said at least one gene following said treating with said compound relative to said level of expression of said at least one gene prior to said treating with said compound indicates that said compound is efficient for treating the subject;

(ii) wherein an increase above a predetermined threshold in said level of expression of said at least one gene following said treating with said compound relative to said level of expression of said at least one gene prior to said treating with said compound indicates that said compound is not efficient for treating the subject; or

(iii) wherein when a level of expression of said at least one gene following said treating with said compound is identical or changed below a predetermined threshold as compared to prior to said treating with said compound then the treatment is not efficient for treating the subject

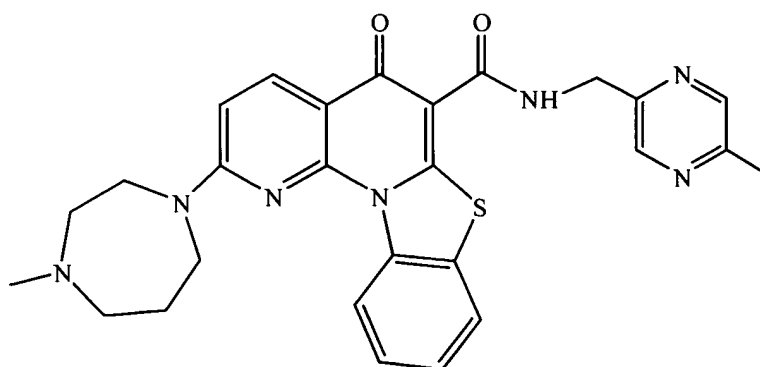
thereby monitoring treatment efficiency of the subject having an autoimmune disease.

11. The method of any of claims 1-10, wherein said compound is 2-(4-methyl-1,4-diazepan-1-yl)-N-((5-methylpyrazin-2-yl)methyl)-5-oxo-5H-benzo[4,5]thiazolo[3,2-a][1,8]naphthyridine-6-carboxamide as depicted in Formula



12. The method of any of claims 1-11, wherein the autoimmune disease is multiple sclerosis.

13. The method of any of claims 1-10, wherein said compound is 2-(4-methyl-1,4-diazepan-1-yl)-N-((5-methylpyrazin-2-yl)methyl)-5-oxo-5H-benzo[4,5]thiazolo[3,2-a][1,8]naphthyridine-6-carboxamide as depicted in Formula



and wherein the autoimmune disease is multiple sclerosis.

14. The method of claims 12 or 13, wherein said multiple sclerosis is a relapsing-remitting course of multiple sclerosis.

15. The method of claims 12 or 13, wherein said multiple sclerosis is a progressive course of multiple sclerosis.

16. The method of claim 12 or 13, wherein said multiple sclerosis is a benign multiple sclerosis.

17. The method of any of claims 1-16, wherein said administering said compound is performed following diagnosis of said autoimmune disease.

18. The method of claim 17, wherein said autoimmune disease is multiple sclerosis and whereas said diagnosis comprises appearance of brain lesions characteristics of said multiple sclerosis.

19. The method of claim 13, wherein said compound is provided at a concentration range of about 0.081-1.62 mg/kg/day.

20. The method of claim 13, wherein said compound is provided at a concentration range of about 0.81-1.215 mg/kg/day.

21. The method of claim 13, wherein said compound is provided at a concentration of about 1.01 mg/kg/day.

22. The method of claim 1, wherein Z^1 is N, and each of Z^2 , Z^3 and Z^4 is C.

23. The method of any of claims 1, 12 and 14-18, wherein U is —W or -L-W, where W is an optionally substituted 5-6 membered unsaturated or aromatic azacyclic ring, optionally containing an additional heteroatom selected from N, O and S; or W is an optionally substituted 5-7 membered saturated azacyclic ring containing an additional heteroatom selected from N and S.

24. The method of any of claims 1, 12 and 14-18, wherein U^2 is $-L-N(R)-W^0$.
25. The method of any of claims 1, 12 and 14-18, wherein Y is an optionally substituted phenyl ring.
26. The method of any of claims 2, 12 and 14-18, wherein at least one of A and X is a tertiary amine A^3 .
27. The method of any of claims 2, 12 and 14-18, wherein A^3 is selected from the group consisting of imidazole, imidazoline, pyrroline, piperidine, piperazine, morpholine, thiomorpholine and homopiperazine.
28. The method of any of claims 2, 12 and 14-18, wherein U^1 is a $-C(=T)N(R)-$, T is O, and U^2 is $-L-W$ or $-L-N(R)-W^0$.
29. The method of any of claims 6, 12 and 14-18, wherein X^1 is CH and two of X^2, X^3, X^4, X^5, X^6 and X^7 are NR^4 .
30. The method of any of claims 6, 12 and 14-18, wherein X^1 is CH and one of X^2, X^3, X^4, X^5, X^6 and X^7 are NR^4 .
31. The method of any of claims 6, 12 and 14-18, wherein X^1 is CH and none of X^2, X^3, X^4, X^5, X^6 and X^7 are NR^4 .
32. The method of any of claims 6, 12 and 14-18, wherein X^1 is N and none of X^2, X^3, X^4, X^5, X^6 and X^7 are NR^4 .
33. The method of any of claims 6, 12 and 14-18, wherein X^1 is N and one of X^4 or X^5 is NR^4 .
34. The method of any of claims 7, 12 and 14-18, wherein R^7 is H and R^8 is

a C₁₋₄ alkyl substituted with an optionally substituted aromatic heterocyclic ring.

35. The method of any of claims 7, 12 and 14-18, wherein the optionally substituted aromatic heterocyclic ring is selected from pyridine, pyrimidine, pyrazine, imidazole, pyrrolidine, and thiazole.

36. The method of any of claims 7, 12 and 14-18, wherein R⁷ and R⁸ together with N in -NR⁷R⁸ form an optionally substituted azacyclic ring selected from the group consisting of morpholine, thiomorpholine, piperidine or piperazine ring.

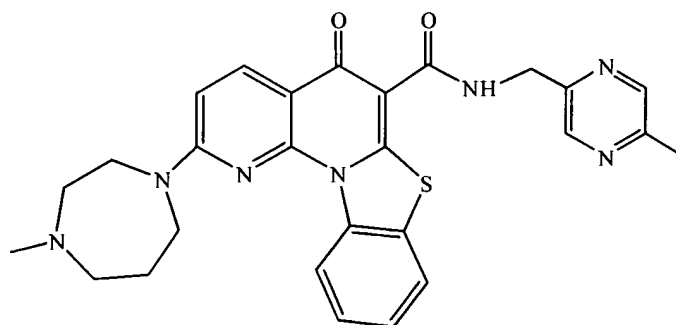
37. The method of any of claims 8, 12 and 14-18, wherein A and V are independently H or halo.

38. The method of any of claims 8, 12 and 14-18, wherein R⁴ is H or C1-4 alkyl.

39. The method of any of claims 8, 12 and 14-18, wherein m and n are each 0.

40. The method of any of claims 8, 12 and 14-18, wherein p is 0 or 1.

41. The method of any of claims 8, 12 and 14-18, wherein said compound having the structure:



or a pharmaceutically acceptable

salt or ester thereof.

42. The method of claim 1, with the proviso that when Z¹ is N, Z² and Z⁴ are

C, Z⁵ is C, U¹ is -C(O)NH-, U² is -L-W, and L is an ethylene linker, one of V, A, and X is independently an optionally substituted aryl, heteroaryl, or 7-membered azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member, if W is pyrrolidin-1-yl, N-methyl-pyrrolidin-2-yl, piperidin-1-yl or morpholin-1-yl.

43. The method of claim 2, with the proviso that when U¹ is -C(O)NH-, U² is -L-W, and L is an ethylene linker, one of A and X is independently an optionally substituted aryl, heteroaryl, or 7-membered azacyclic ring, optionally containing an additional heteroatom selected from N, O and S as a ring member, if W is pyrrolidin-1-yl, N-methyl-pyrrolidin-2-yl, piperidin-1-yl or morpholin-1-yl.